THE FINITE-PRESENTATION QUOTIENT FUNCTOR IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-q (canonical tag 082M). cited at statement line 2164. Standing situation: f : X -> B with the finite presentation quotient functor Q^fp_{F/X/B}.
- Lemma spaces-limits-lemma-directed-inverse-system-has-limit (canonical tag 07SF). cited at proof line 2181. X_T = lim X_{T_i} for the directed limit of affine schemes.
- Lemma spaces-limits-lemma-descend-modules-finite-presentation (canonical tag 07V7). cited at proof line 2185. The quotient map F_T -> Q descends to F_{T_i} -> Q_i of finite presentation.
- Lemma spaces-properties-lemma-quasi-compact-affine-cover (canonical tag 03H6). cited at proof line 2195. Affine etale cover U -> X reduces surjectivity and flatness checks to the affine case.
- Lemma algebra-lemma-module-map-property-in-colimit (canonical tag 05LI). cited at proof line 2208. Surjectivity at the colimit holds already at a finite stage.
- Lemma algebra-lemma-flat-finite-presentation-limit-flat (canonical tag 02JO). cited at proof line 2210. Flatness at the colimit holds already at a finite stage.

Source reference. Original definition 082M: sources/stacks-chow/tags.txt, source lines 10257–10257. Cited at statement line 2164: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 07SF: sources/stacks-chow/tags.txt, source lines 9900–9900. Cited at proof line 2181: the base change X_T is the directed limit of the X_{T_i}; load-bearing.

Source reference. Original lemma 07V7: sources/stacks-chow/tags.txt, source lines 9997–9997. Cited at proof line 2185: the quotient map F_T -> Q descends to some F_{T_i} -> Q_i of finite presentation; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Cited at proof line 2195: choose the affine etale cover U -> X to reduce to the affine case; load-bearing.

Source reference. Original lemma 05LI: sources/stacks-chow/tags.txt, source lines 7237–7237. Cited at proof line 2208: surjectivity of M_i -> N_i holds after enlarging i; load-bearing.

Source reference. Original lemma 02JO: sources/stacks-chow/tags.txt, source lines 3343–3343. Cited at proof line 2210: flatness of N over A holds after enlarging i; load-bearing.

Lemma 0.2. *In Situation 082M assume also that (a) f is quasi-compact and quasi-separated and (b) $\mathcal{F}$ is of finite presentation. Then the functor $Q^{fp}_{\mathcal{F}/X/B}$ is limit preserving in the following sense: If $T = \lim T_i$ is a directed limit of affine schemes over B, then $Q^{fp}_{\mathcal{F}/X/B}(T) = \operatorname{colim} Q^{fp}_{\mathcal{F}/X/B}(T_i)$.*

Proof. Let $T = \lim T_i$ be as in the statement of the lemma. Choose $i_0 \in I$ and replace I by $\{i \in I \mid i \geq i_0\}$. We may set $B = S = T_{i_0}$ and we may replace X by X_{T_0} and $\mathcal{F}$ by the pullback to X_{T_0}. Then $X_T = \lim X_{T_i}$, see Limits of Spaces, Lemma 07SF. Let $\mathcal{F}_T \to \mathcal{Q}$ be an element of $\mathrm{Q}^{fp}_{\mathcal{F}/X/B}(T)$. By Limits of Spaces, Lemma 07V7 there exists an i and a map $\mathcal{F}_{T_i} \to \mathcal{Q}_i$ of $\mathcal{O}_{X_{T_i}}$-modules of finite presentation whose pullback to X_T is the given quotient map.

We still have to check that, after possibly increasing i, the map $\mathcal{F}_{T_i} \to \mathcal{Q}_i$ is surjective and $\mathcal{Q}_i$ is flat over T_i. To do this, choose an affine scheme U and a surjective étale morphism $U \to X$ (see Properties of Spaces, Lemma 03H6). We may check surjectivity and flatness over T_i after pulling back to the étale cover $U_{T_i} \to X_{T_i}$ (by definition). This reduces us to the case where $X = \operatorname{Spec}(B_0)$ is an affine scheme of finite presentation over $B = S = T_0 = \operatorname{Spec}(A_0)$. Writing $T_i = \operatorname{Spec}(A_i)$, then $T = \operatorname{Spec}(A)$ with $A = \operatorname{colim} A_i$ we have reached the following algebra problem. Let $M_i \to N_i$ be a map of finitely presented $B_0 \otimes_{A_0} A_i$-modules such that $M_i \otimes_{A_i} A \to N_i \otimes_{A_i} A$ is surjective and $N_i \otimes_{A_i} A$ is flat over A. Show that for some $i' \geq i$ $M_i \otimes_{A_i} A_{i'} \to N_i \otimes_{A_i} A_{i'}$ is surjective and $N_i \otimes_{A_i} A_{i'}$ is flat over A. The first follows from Algebra, Lemma 05LI and the second from Algebra, Lemma 02JO. $\square$